

C

DHTSystem

package dht0

import java.util.ArrayList

import java.util.List

void addNode(NodeThread node)

nodes.add(node);

node.start();

void buildDiscoveryChain()

void printChain()

C

Main

package dht0

static void main(String[] args) throws InterruptedException

system.addNode(n1);

system.addNode(n2);

system.addNode(n3);

system.buildDiscoveryChain();

system.printChain();

n1.receiveMessage(putMsg);

Thread.sleep(500);

n3.receiveMessage(getMsg);

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Message

package dht0

Type type

String key

String value

Node origin

I

Node

package dht0

void receiveMessage(Message msg);

long getId();

C

NodeThread

package dht0

import java.util.\*

import java.util.concurrent.\*

final long id

NodeThread(long id)

void run()

processMessage(msg);

} catch (InterruptedException e)

Thread.currentThread().interrupt();

void processMessage(Message msg)

switch (msg.type)

localStore.put(msg.key, msg.value);

forwardToPeers(msg);

void forwardToPeers(Message msg)

for (Node peer : peers)

peer.receiveMessage(msg);

void addPeer(Node peer)

void receiveMessage(Message msg)

mailbox.offer(msg);

synchronized (System.out)